Neutropenic Fever

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35.1 Introduction

Fever during neutropenia is almost universal after an HCT. In neutropenic HCT recipients, clinicians are faced with a unique combination of issues: (1) high incidence of bacterial bloodstream infections, (2) high mortality in case of infections due to Gram-negative bacteria unless effective antibiotic treatment is provided promptly, and (3) numerous other causes of fever.

Additionally, in the absence of neutrophils which are responsible for most of clinical signs or symptoms during a localised bacterial infection (abscess formation, prominent lung infiltrates, pyuria, etc.), fever is frequently the only symptom present also in these cases. On the other hand, fever is a highly unspecific sign, and there are numerous causes of fever during neutropenia other than bacterial infections, including (a) viral infections, (b) fungal infections, (c) drug reactions (e.g. ATG), (d) transfusion reactions, (e) cytokine release syndrome (CRS), (f) mucositis (g) engraftment syndrome, (h) GvHD, (i) rejection, (j) haemophagocytosis, and (k) underlying disease.

However, since infection due to Gram-negative bacteria, particularly *Pseudomonas aeruginosa*, can result in rapid deterioration of clinical conditions and death, this possibility should always be considered and appropriate empirical antibiotic therapy started while awaiting the results pointing to the actual cause of fever or absence of bloodstream infection. The issue of prevention of fever and infections during neutropenia through antibiotic prophylaxis with fluoroquinolones has been seriously challenged by a worldwide increase in antibiotic resistance (Mikulska et al. 2018).

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35.2 Initial Management of Fever During Neutropenia

Initial management of fever during neutropenia should include all the following (Freifeld et al. 2011; Averbuch et al. 2013; Lehrnbecher et al. 2023).

35.2.1 Diagnostic Procedures

(a)	At least two sets (1 set = 1 aerobic and 1 anaerobic bottle) of **blood cultures**
1.	Including at least one set from the central venous catheter (CVC), if present
2.	Using an aseptic methodology to reduce the risk of contamination
3.	Providing adequate blood volume (10 ml in each bottle), since the volume of blood is essential to ensure optimal detection of bacteraemia or candidemia

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(b) **Clinical exam** with particular attention to subtle signs of a localised infection
   4. Signs of infection of exit/entry of CVC
   5. Perineal pain suggestive of an abscess
   6. Skin or nail lesions suggestive of fungal infection
   7. Abdominal defence or diarrhoea suggestive of neutropenic enterocolitis, C. difficile infection, etc.
   8. Upper respiratory tract symptoms such as rhinorrhoea suggestive of viral respiratory infection
   9. Mucosal lesions
   10. CNS signs or symptoms (focal lesions, e.g. with fungal infection or bacteria abscess vs. being confused in severe systemic infection or viral encephalitis)
   11. Pleuritic chest pain suggestive of invasive mould infection
(c) Any **other microbiological tests** based on the clinical presentation (e.g. sputum culture, nasal swab for respiratory viruses, urinary antigen for *legionella*, viral DNA (CMV, ADV, etc.), *Clostridioides difficile* toxin, etc)
(d) **Radiological tests** based on the clinical presentation (for suspected lung involvement, lung CT should be used since chest X-ray has too low sensitivity for detecting pneumonia in neutropenic patients) and the previous episodes of fever

35.2.2 Evaluation of the Risk of Clinically Severe Infection

Such an evaluation, based on comorbidities and clinical presentation, leads to the decision on additional diagnostic tests, the spectrum of antibiotic coverage, the need for close monitoring for sings of further clinical deterioration, and, in case of outpatients, on hospital admission.

35.2.3 Evaluation of the Risk of Infection Due to Resistant Bacteria (Particularly Gram-Negative)

This risk is considered high in case of:

(a) Colonisation with a resistant bacterial strain.
(b) Previous infection caused by a resistant bacterial strain.
(c) Local epidemiology with high incidence of infections caused by resistant pathogens.

35.2.4 Choice of the Appropriate Empirical Antibiotic Therapy

In order to provide immediate active treatment and prevent overexposure to broad spectrum agents/combinations, the choice of escalation and de-escalation strategy (see Table 35.1) and the most appropriate antibiotic must be made.

35.2.5 In High-Risk Patient's Assessment of the Need for Antifungal Therapy

(a) Assess the risk of candidemia in patients not receiving antifungal prophylaxis and presenting with septic shock.
(b) Assess the probability of invasive aspergillosis (IA) based on the incidence of IA (taking into account risk factors, mould-active prophylaxis, etc.) and the results of galactomannan (GM) or *Aspergillus* PCR screening or targeted testing.

Empirical antifungal therapy (adding antifungal agent in patients persistently febrile despite broad-spectrum antibiotics) could be replaced by diagnostic-driven strategy based on the use of diagnostics, such as lung CT, fungal serum markers (GM, PCR, β-D-glucan) and targeted treatment following diagnosis (see Chap. 37).

35.3 Main Changes in the Management of Neutropenic Fever

The main change in the management of febrile neutropenia occurred due to an increasing rate of *multidrug-resistant (MDR) bacteria* in certain countries or centres, in particular Gram-negative rods resistant to almost all antibiotics routinely

Table 35.1 The main characteristics of escalation and de-escalation strategy

Strategy	Escalation	De-escalation
Definition	Empirical treatment active against susceptible *Enterobacteriaceae* and *P. aeruginosa*	Starting upfront an empirical coverage of **MDR** bacteria, particularly **Gram-negative**, which is ***later (72–96 h) reduced (de-escalated)*** if a MDR pathogen is *not* isolated, i.e.: • Susceptible strain isolated • No microbiological results
Antibiotics usually used	Monotherapy with anti-pseudomonal cephalosporin (cefepime, ceftazidime) or piperacillin-tazobactam	• Carbapenem or a new combination of β-lactam/β-lactam-inhibitor (e.g. ceftolozane/tazobactam or ceftazidime/avibactam), to cover ESBL producers and some resistant *P. aeruginosa* strains Combinations, examples • β-lactam + aminoglycoside • β-lactam + coverage of resistant Gram-positives Optimal choice will depend on colonising MDR strain or local epidemiology
Main advantages	Less induction or selection of resistant strains (carbapenem sparing) Less toxicity	Appropriate therapy before culture results are available > **lower mortality**
Main limitations	In case of empirical therapy inactive against isolated Gram-negative, prognosis is significantly worsened	Overuse of broad-spectrum antibiotics/combinations leading to high antibiotic pressure, particularly in case of failure to de-escalate
Who	All patients, unless criteria for de-escalation approach are present	Patients at risk for infections due to resistant bacteria, such as those with: • Colonisation with a resistant pathogen • Previous infection with a resistant pathogen • Cared for in centres in which resistant pathogens are frequently isolated Particularly if presenting in severe clinical conditions

MDR multidrug resistant

used in febrile neutropenia (e.g. *Enterobacteriaceae* resistant to third-generation cephalosporins ± piperacillin-tazobactam, i.e. producers of extended-spectrum β-lactamases [ESBLs]; *Enterobacteriaceae* or *Pseudomonas aeruginosa* or *Acinetobacter baumannii* resistant to carbapenems). Therefore, traditional protocols consisting of starting a standard β-lactam in all febrile patients and changing treatment in case of persistent (48–72 h) fever (called escalation strategy) might not be applicable to all patients, and individualised approach might be required.

35.3.1 De-escalation Strategy

Patients who are at high risk of infections due to resistant bacteria (particularly Gram-negative) should immediately receive treatment with antibiotics active against such strain since the delay in starting effective antimicrobial therapy has been associated with an increased mortality (Averbuch et al. 2017). Therefore, a de-escalation strategy, typically used in critically ill patients in intensive care units (ICU), has also been proposed for neutropenic haematology patients (Averbuch et al. 2013).

Traditional *escalation empirical therapy* (Table 35.1) is still appropriate in most of cases, especially in countries or centres when resistance rates are low among pathogens commonly causing infections in neutropenia. With this approach, we avoid routine use of carbapenems or combinations of a β-lactam with aminoglycoside (which have been shown associated with more toxicity and no clinical advantage) (Averbuch et al. 2013; Drgona

et al. 2007). The empirical use of an antibiotic active against resistant Gram-positive bacteria (such as vancomycin or agents active against vancomycin-resistant enterococci) is not recommended neither as initial therapy nor in persistently febrile patients, unless the patient has signs or symptoms suggesting a Gram-positive aetiology (e.g. skin or CVC involvement or pneumonia) or a documented Gram-positive infection (Freifeld et al. 2011; Beyar-Katz et al. 2017; Kamboj et al. 2019).

De-escalation strategy consists of starting with a broad initial empirical regimen, chosen due to on the severity of the patient's clinical presentation and the risk of infection due to resistant (mainly Gram-negative) bacteria based on individual factors for harbouring MDR bacteria and the local bacterial epidemiology. The key issues of de-escalation approach are (1) providing immediate effective treatment of a potentially life-threatening MDR pathogen and (2) reducing as much as possible the unnecessary use of broad-spectrum drugs (to avoid selection or induction of resistant strains). Data from neutropenic cancer patients in ICU and HCT recipients showed that de-escalation approach is safe and feasible (Mokart et al. 2014; Snyder et al. 2017; Gustinetti et al. 2018), but its implementation is not universal HCT centres (Verlinden et al. 2020). Main characteristics of escalation and de-escalation approach are reported in Table 35.1. The choice of antibiotics for de-escalation strategy will depend on susceptibility results in case of previous infection or colonisation with MDR Gram-negative (in that case empirical treatment can be decided as soon as these results are available, i.e. before the development of fever and indicated in the clinical chart to be started in case of fever) or local epidemiology, e.g. high prevalence of extended-spectrum beta-lactamases (ESBL)-producing Enterobacteriaceae in bloodstream isolates.

35.3.2 Antibiotic Discontinuation

Another issue of management of febrile neutropenia is the *length of antibiotic therapy*, particularly in the absence of clinically or microbiologically documented infection. Traditionally, antibiotic treatment was continued until neutrophil recovery, with the aim of avoiding infection relapse. This approach has been challenged by IDSA and ECIL guidelines, with the latter stating that antibiotics can be safely discontinued after ≥72 h of IV therapy in patients that are and have been haemodynamically stable since the onset of fever and are afebrile for ≥48 h, irrespective of the granulocyte count and the expected duration of neutropenia (Averbuch et al. 2013; Lehrnbecher et al. 2023). The rational for this recommendation was the fact that alteration of patient's microbiota leads to an increased risk of colonisation/selection of resistant pathogens, which might subsequently cause life-threatening infections.

The safety of *discontinuation of empirical antibiotic* therapy after few days of treatment, provided the antibiotic treatment is restarted immediately in case of fever reappearance, has been reported and demonstrated in several studies (Orasch et al. 2015). A randomised open-label trial performed in 157 high-risk febrile neutropenic haematology patients without clinically or microbiologically documented infection showed that antibiotics can be safely discontinued after 72 h of apyrexia and clinical recovery, irrespective of the neutrophils count, saving days of antibiotic exposure (Aguilar-Guisado et al. 2017). Another randomised trial aimed to validate a clinical practice of discontinuing empirical carbapenem therapy of febrile neutropenia after 3 days, irrespective of fever resolution, as long as no clinically or microbiologically documented infection was present (de Jonge et al. 2022). In that study, most (>60%) of 281 included patients were autologous transplant recipients. There was no difference in treatment failure and no deaths due to carbapenem-susceptible bacteria, but serious adverse events were more frequent in short treatment group (mainly readmissions and mainly in those persistently febrile), suggesting that obtaining clinical improvement before discontinuation might be required.

35.4 Fever Persistent despite Empirical Antibiotic Therapy

Fever persistent despite empirical antibiotic therapy is not an infrequent event. Patient's general clinical conditions are the most important factor to consider.

If no signs or symptoms of clinical deterioration (e.g. septic shock, confusion, worsening respiratory function) are present, slow response to antibiotic treatment should be considered, particularly if accompanied by improvement in inflammatory markers such as C-reactive protein or procalcitonin (the latter particularly for Gram-negative bloodstream infections). In alternative, nonbacterial infections (e.g. viral or fungal) or noninfectious causes, such as mucositis, CRS or engraftment syndrome, should be considered. Usually, changes in antibiotic regimen are not necessary if clinical conditions are stable. Routine addition of antibiotics against resistant Gram-positives (glycopeptides) has not been shown effective (Beyar-Katz et al. 2017).

Results of serum GM, PCR or other fungal markers, performed either in screening or at the onset of fever, should be available by day 2–3 of fever and should guide antifungal treatment. In patients at high risk of IA, lung CT scan should be performed. *Empirical antifungal treatment* has been introduced when noninvasive diagnostic tests were not available and CT availability was extremely limited. When these diagnostic measures became available, *pre-emptive (called also diagnostic-driven) approach* has been shown able to provide earlier treatment than empirical approach (Maertens et al. 2005) (see Chap. 37). Empirical antifungals might be provided while awaiting the results of diagnostic tests or, in case of mould-active prophylaxis, the confirmation of adequate blood levels, but appropriate diagnostics should be performed to confirm or exclude the presence of invasive fungal disease. In a randomised trial, pre-emptive antifungal strategy was safe in high-risk neutropenic patients on fluconazole prophylaxis, and reduced by half the number of patients receiving antifungals without excess mortality or increase in invasive fungal infections (Maertens et al. 2023).

If clinical conditions deteriorate, usual management steps are:

1. *Aggressive diagnostic workup* (repeated blood cultures, additional testing for viruses and fungi, CT scan, BAL lavage in case of pneumonia, lumbar puncture in case of CNS symptoms, etc.), and while awaiting the results:
2. *Escalation* of antibacterial treatment
3. In some cases starting an *antifungal therapy.*

There is no universal scheme for antibiotic escalation therapy, but it usually covers resistant Gram-negatives (including those producing ESBLs, e.g. with a carbapenem or an addition of aminoglycoside) and methicillin-resistant staphylococci or ampicillin-resistant enterococci (e.g. with a vancomycin or novel agents). Coverage of other resistant bacteria should be based on the local epidemiology, the epidemiology of a centre where the patient was cared for before transplant and on patient's past history of infections and colonisation. In case of diarrhoea, diagnosis of *Clostridioides difficile* infection is mandatory since antibiotic escalation without appropriate therapy for *C. difficile* might worsen the infection. Nonbacterial infections (viral, fungal, toxoplasma, etc) and less frequent agents (e.g. Legionella, mycobacteria, or *Nocardia*) should be considered in differential diagnosis and tested for based on clinical presentation and patient's past exposure. Empirical antifungal treatment in this setting might be warranted while awaiting the results of the whole diagnostic workup.

Key Points

- Numerous causes of fever during neutropenia exist, but usually neutropenic fever should be managed as suspected bloodstream infection due to Gram-negatives until proven otherwise.
- The initial management includes diagnostics (two sets of blood cultures) and the assessment of the risk of (1) clinically severe infection and (2) infection due to resistant bacteria.
- In patients with severe presentation and the risk of resistant bacteria, de-escalation approach should be used in order to cover the most probable resistant strain(s).
- In other cases, escalation approach is appropriate and the choice of the first-line empirical antibiotic therapy should be based on antibiotic susceptibility of Gram-negative bacteria most frequently isolated in one's centre.
- Empirical antifungal therapy could be replaced in most cases by diagnostic-driven (pre-emptive) strategy.
- In the absence of clinically or microbiologically documented infection, empirical antibiotic can be safely discontinued after 72 h of apyrexia and clinical recovery, irrespective of the neutrophils count, and it saves exposure to antimicrobials.
- In case of clinical worsening and persistence of fever, extensive diagnostic workup is mandatory.

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